

Jayesh Kumar Singh

Full-Stack AI Engineer · GenAI · RAG · Agentic Systems (MCP) · Node.js/.NET/React

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SUMMARY

Full-stack **AI engineer** with **3+ years** shipping production **RAG** and **agentic AI** systems (**MCP** tool servers, **LLM** orchestration, **embeddings**, **vector search**) on top of enterprise full-stack platforms (**Node.js**, **.NET**, **React/Angular**). Cut document review time **~70%** and repeat support queries **~30%** by shipping a grounded RAG assistant into a live Government of India PSU system; sustained **~99% database uptime**; delivered **3 production modules** with zero missed release deadlines.

EXPERIENCE

Utility Powertech Limited

Aug. 2026 - Present

Consultant -- IT (Full-Stack AI Engineer)

Noida, India

- Own **prompt engineering** end-to-end for **OpenAI/Claude** integrations on a client-facing product feature - designing, testing, and iterating templates until output is consistently reliable and production-ready.
- Integrated an **LLM-powered** assistant into **.NET (C#)** backend services, reducing manual review effort needed before output reached users, from prompt design through API integration.
- Built and shipped **Angular** UI components consuming **.NET REST APIs** for core client-facing screens, working directly with product/design to ship iteratively.
- Contributed **React-based** components extending the product's AI-assisted feature set, working across both **Angular** and **React** front-ends on the same platform.

GA Digital Web Word Pvt. Ltd.

Apr. 2026 - Aug. 2026

Consultant -- IT (Full-Stack Engineer, Contract)

Gurugram, India

- Cut repeat support queries **~30%** by building a **RAG** layer over internal documents (**embeddings** + **vector search**) exposed via **REST APIs**, tuning chunking against measured retrieval hit-rates.
- Cut document lookup and review time **~70%** by shipping a **multi-agent LLM** (OpenAI/Claude) document-search and summarization assistant into the **Node.js/.NET** stack, with prompt templating, output validation and **RBAC-gated** access.
- Delivered **3 production modules** end-to-end (requirements → deploy) on **Node.js/Express**, **.NET (C#)** and **React/Angular** for **Energy Efficiency Services Limited (EESL)**, a Government of India PSU, with zero missed release deadlines using **AWS**, **Docker** and **GitHub Actions CI/CD**.
- Designed **6+ JWT-secured REST endpoints** per module over **MySQL**, **MSSQL** and **PostgreSQL**, paired with reusable **React/Angular** components.

SISL Infotech Pvt. Ltd.

Feb. 2025 - Mar. 2026

Consultant -- IT (Full-Stack Engineer)

Noida, India

- Cut notification failures **~90%** (to near-zero) by building **Dockerized** email-notification microservices with retry logic, health monitoring and structured logging.
- Cut manual compliance-review time **~60%** by shipping **LLM-powered** auto-summarization of compliance findings and notification drafting, wired into the production email-microservice pipeline.
- Resolved **100+ compliance findings** across **15+ enterprise site audits**, building **React/Angular** dashboards over **Node.js/.NET (C#) REST APIs** backed by **MySQL/MSSQL**.
- Cut slow-query times **~70%** through **MySQL/MSSQL** tuning and indexing; led schema design and data migration across **MySQL**, **MSSQL** and **PostgreSQL**.
- Secured **4 enterprise modules** with **JWT** + **Azure AD RBAC** across **Node.js** and **.NET (C#) REST APIs**; delivered a compliance/approval system (SLA tracking, audit logs) and a role-based **Bill Tracking System**.
- Prototyped **PyTorch** embeddings and small neural networks on compliance-document data to evaluate model behavior for future AI features.

iONE IT Solution Pvt. Ltd.

May 2024 - Jan. 2025

Database Engineer

Noida, India

- Sustained **~99% uptime** engineering production database **HA**: schema design, indexing, backups, **point-in-time recovery** and standby replication.
- Cut query times **~70%** via **EXPLAIN ANALYZE**-driven tuning and indexing; automated backup, recovery and **migration** pipelines across environments - the same performance fundamentals that drive **vector-store** retrieval speed.
- Applied **scikit-learn** to production database telemetry for anomaly detection and query-performance forecasting, building **feature pipelines** that fed downstream **ML** and **vector-search** workloads.

AAL Infotech Pvt. Ltd.

Backend Engineer

Jul. 2023 - Apr. 2024

India

- Improved API response times **~60%** building **Node.js/Express REST APIs** over **MongoDB** with schema and **aggregation-pipeline** design.
- Built backend **data pipelines** and integrated **AI/LLM APIs** into **Node.js/Express** endpoints to power AI-driven product features.

PROJECTS

Healthcare AI Agent Platform - Agentic AI over Medical Knowledge | *MCP, RAG, LLMs, pgvector, Next.js, Python* 2026

- Exposed **36+ tools** to **LLM agents** over the **Model Context Protocol (MCP)**, enabling multi-step clinical-education workflows through structured tool calls instead of free-text prompts.
- Built a **RAG** pipeline over **37 medical textbooks** - chunking, **Transformers.js embeddings** and **vector search** on **PostgreSQL + pgvector** - to ground agent responses in source material and reduce hallucination.
- Shipped **dual MCP transports (stdio + Streamable HTTP)** plus a mirrored **REST API**, letting the same tool layer serve local agents, remote clients and standard web integrations.
- Built the product as **Next.js (React)** with a **Python Flask** ML sidecar and **Prisma**-backed state, containerized end-to-end with **Docker**; delivered as a **team of 2**, owning agent/tool design, retrieval quality and deployment.

eBegin - Enterprise Management Platform | *React, Node, Express, MySQL/MSSQL*

2025

- Unified **SCM, HCM, CCMS** and **Finance** modules on a **React** front-end and **Node.js/Express** APIs, with **JWT + Azure AD RBAC** and optimized **MySQL/MSSQL** stored procedures.

FoodZoned - Full-Stack Food Delivery App | *React, Node, Express, MongoDB*

2023

- Shipped a full-stack ordering app with **JWT** auth, **Razorpay** payment integration and automated email confirmations.

SKILLS

AI / GenAI : LLMs (OpenAI, Claude), Agentic Systems, Model Context Protocol (MCP), RAG, Embeddings, Vector Search (pgvector), Transformers.js, Prompt Engineering, Tool / Function Calling

ML / Data : Python, scikit-learn, PyTorch (fundamentals), Flask, Pandas, NumPy, Statistics & Data Analysis, Data Pipelines

Full-Stack : Node.js, Express.js, .NET (C#), React.js, Next.js, Angular, TypeScript, JavaScript (ES6+), Redux, Tailwind, REST APIs, Microservices, JWT, RBAC

Databases : PostgreSQL + pgvector, MongoDB, MySQL, MSSQL, Prisma, Mongoose

Cloud / DevOps : AWS, Docker, CI/CD (GitHub Actions), Git, Linux

EDUCATION

Dr. Akhilesh Das Gupta Institute of Technology and Management

B.Tech, Information Technology (CGPA: 8.6)

2019 - 2023

Delhi, India

CERTIFICATIONS

DSA & OOP - takeUforward (2026) | **Machine Learning A-Z** (Python & R) - Udemy (2023) | **React** - Internshala (2023) | **Web Development** - Internshala (2021)